Manual Calculation

EXP-1

1) program -A

8- bit addition and subtraction

given A = 5AH, B = 27H

Addition :-

5AH + 27H = 81H

Sum = 81H, Cy = 0

AC = 1, P = 1, Z = 0, S = 1

Flag byte = 94H

Signed overflow = yes

Subtraction :-

5AH - 27H = 33H

Diff = 33H,

Borrow = 0, AC = 0,

P = 1, Z = 0, S = 0

Flag byte = 06H

Signed overflow = No

2) program-B

16 bit addition and subtraction

given X = 1234H, Y = 4321H

Addition :-

1234H + 4321H = 5555H

Carry = 0

Subtraction :-

1234 - 4321H = CF13

Borrow = 1

3) Program-C :-

8- bit Multiplication 14H x 06H

Division F2H ÷ 0AH

Multiply :

14H x 06H

= 0678H

Division

F2H ÷ 0AH

Quo = 18H

Rem = 02H

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Program - D

16 bit multiplication & division

given

multiplication : 1357H × 0025H

Division : 9A23H ÷ 0011H

Multiply

1357H × 0025H = 0002CA13H

Upper word = 0002H

Lower word = CA13H

Division :

9A23H ÷ 0011H

Quo = 0911H

Rem = 0002H

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